

# SHWETA KUMARI

Senior Backend Engineer | Java, Spring Boot & Distributed Systems | 7+ Years

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## PROFESSIONAL SUMMARY

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Senior Backend Engineer with 7+ years designing and shipping high-throughput microservices on Java (8–21), Spring Boot, Apache Kafka, and Solace JMS. Specialized in secure, low-latency systems — mTLS architectures, JWT-based authentication, and multi-zone integrations — supporting millions of daily transactions in telecom and retail domains. Strong in system design, concurrency, performance tuning, and observability (ELK Stack) across Linux and AWS environments. Recently shipped a full-stack, containerized side project deployed end-to-end on AWS EC2.

## TECHNICAL SKILLS

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**Languages:** Java (8, 11, 17, 21), SQL, C, HTML, XML, JSON, YAML

**Frameworks:** Spring Boot 3, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, SOAP APIs, React 18, Vite, Tailwind CSS

**Architecture & Design:** Microservices, Event-Driven Architecture, System Design, OOP, SOLID Principles, Design Patterns (Singleton, Factory, Builder, Strategy, Observer), API Design, High Availability, Low-Latency Systems, Scalability

**Messaging & Streaming:** Apache Kafka, Solace JMS, Pub/Sub, Message Queues

**Security:** mTLS (Mutual TLS), JWT, OAuth 2.0 concepts, RBAC, Spring Security, SSL/TLS, X.509 Certificates, Password Hashing (BCrypt)

**Cloud & DevOps:** AWS (EC2, S3, IAM, Security Groups), Docker, Docker Compose, Multi-stage Dockerfiles, Jenkins, CI/CD Pipelines, Git, GitHub, Bitbucket, NGINX (Reverse Proxy), API Gateway, Linux/Unix

**Observability:** ELK Stack (Elasticsearch, Logstash, Kibana), Structured Logging, Log Analysis, Service Dashboards

**AI Developer Tools:** Claude, ChatGPT, GitHub Copilot

**Concurrency:** Multithreading, ExecutorService, ReentrantLock, CompletableFuture, Thread Pools, Synchronization, Thread Safety

**Databases:** Relational SQL, Stored Procedures, Indexing, Query Optimization, Transactions, ACID, H2

**Testing:** JUnit 5, Mockito, Unit Testing, Integration Testing, TDD

**Build & Tools:** Maven, Gradle, Postman, Swagger/OpenAPI, Jira, IntelliJ IDEA, Eclipse

## WORK EXPERIENCE

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Senior Software Engineer | Java Developer | Bharti Airtel, Gurugram

Mar 2022 – Present

*Postpaid Domain — Backend Architecture & Platform Engineering*

- **mTLS Authentication Service:** Designed a standalone Spring Boot JAR implementing X.509 two-way SSL handshake with custom keystore/truststore lifecycle management — overcoming WebLogic's native mTLS limitation to secure service-to-service communication.
- **BSL Integration Architecture:** Architected a Solace JMS topology with circle-wise queue partitioning and direct pub/sub routing, eliminating middleware hops and cutting API latency by 50% across multi-zone deployments.
- **High-Throughput API Layer:** Built synchronous and asynchronous REST endpoints on Spring Boot using Java 17 Streams, Optional, and CompletableFuture pipelines — sustaining millions of concurrent postpaid user requests.
- **Application Security Stack:** Implemented Spring Security with JWT issuance/validation, RBAC authorization, BCrypt password hashing, and SSL/TLS termination — aligned with enterprise compliance standards.
- **Concurrency & Throughput Tuning:** Redesigned compute-heavy modules with ExecutorService, ReentrantLock, and CompletableFuture chaining — improving throughput by 25% under peak load.
- **Java 8 → Java 21 Migration:** Led runtime and dependency migration across 5+ microservices; adopted records, pattern matching, switch expressions, and virtual threads for better performance and maintainability.
- **Observability Pipeline:** Rolled out ELK Stack (Elasticsearch, Logstash, Kibana) with structured JSON logging and correlation IDs — built Kibana dashboards that reduced MTTR on production incidents.
- **AWS S3 Integration:** Built secure object-storage flows using pre-signed URLs, bucket policies, and lifecycle rules for document upload/retrieval across backend services.

- **Edge & Traffic Layer:** Configured NGINX and API Gateway as reverse proxy/load balancer with health checks and failover routing — improving resilience under high concurrency.
- **CI/CD & Containerization:** Maintained Jenkins pipelines and Dockerized deployments; standardized Git branching strategy on Bitbucket/GitHub for faster, safer releases.
- **Developer Productivity & AI Tooling:** Drove org-wide adoption of GitHub Copilot, Claude, and ChatGPT for code generation, refactoring, and test scaffolding — scaled to 10+ engineers.
- **Quality & Mentorship:** Established review standards and quality gates; mentored peers on SOLID, design patterns, and modern Java — reducing defect density by 15%.

## Software Engineer | Tata Consultancy Services, Kolkata

Jun 2019 – Mar 2022

*Client: Walgreens — Pharmacy & Retail Systems*

- **Backend Services:** Developed and maintained Spring Boot and Spring MVC services on Java 8 powering Walgreens' pharmacy and retail platforms — supporting millions of daily transactions.
- **Database Optimization:** Tuned complex SQL queries and stored procedures for inventory and transaction workflows — reducing query response times by up to 30%.
- **Testing & Quality:** Achieved 90%+ code coverage with JUnit and Mockito across unit and integration tests — driving defect-free production releases.
- **Agile Delivery:** Owned stories across Agile/Scrum sprints in Jira — contributing to estimations, planning, and retrospectives for on-time feature delivery.

## PERSONAL PROJECTS

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### Meetup — Event Management Platform

[Live Demo](#)

*Stack: Java 17, Spring Boot 3, Spring Security, JWT, JPA/Hibernate, H2 | React 18, Vite, Tailwind CSS | Docker, Docker Compose, Nginx, AWS EC2*

- Shipped a production-ready, full-stack event management web app **end-to-end in 3 days** as a sole contributor — owning backend, frontend, containerization, and cloud deployment.
- Designed a **role-based access control (RBAC)** system with two distinct user journeys (Admin / Customer) on Spring Boot 3 + Spring Security, using stateless JWT authentication and an approval-gated registration workflow persisted via JPA/Hibernate on H2.
- Built a responsive **React 18 + Vite + Tailwind CSS** frontend with protected client-side routes, token-based session handling, and a polished approval UX for admin review of pending users.
- Containerized both backend and frontend with **multi-stage Dockerfiles** (slim runtime images, separated build/runtime layers) and orchestrated the stack via **Docker Compose** behind an **Nginx reverse proxy** handling routing and static asset delivery.
- Deployed to **AWS EC2 (free tier)** — provisioned the instance, hardened ingress via Security Groups, installed the Docker runtime, and shipped a publicly accessible live build, demonstrating end-to-end ownership from code to cloud.

## EDUCATION

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### B.Tech — Electronics & Communication Engineering

Jul 2015 – Jun 2019

*Haldia Institute of Technology*

## AWARDS & RECOGNITION

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- **Execution Excellence Award — Bharti Airtel:** For end-to-end BSL system integration and Inflight Flow architecture design.
- **Kudos Award (Great Team Player):** Received at both Bharti Airtel and Tata Consultancy Services for cross-team collaboration and delivery impact.

## COMPETITIVE PROGRAMMING

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**LeetCode:** shweta\_sher (170+ problems solved) | **HackerRank:** shwetajaiswalof1 | **CodeChef:** shweta\_sher